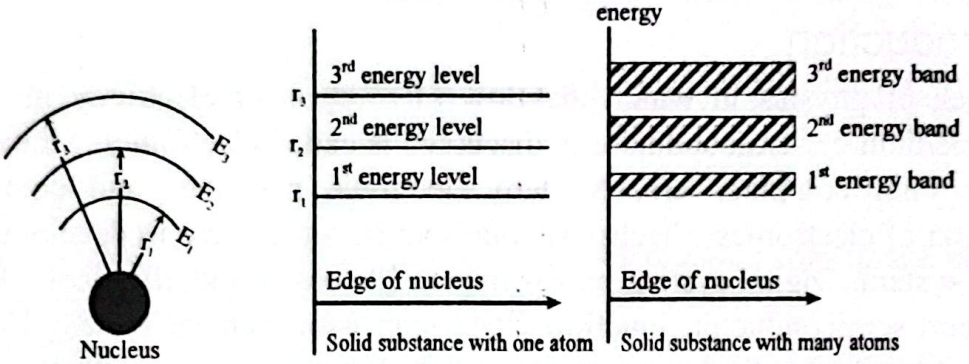
10.1 Band Theory

According to the theory of Neils Bohr, the electron of hydrogen atom can not be energetic of any kind. Because electron can stay in a shell of specific energy. Electrons of any atom revolves round the nucleus in some specific allowed orbits which have specific energy. Electron's energy depends on the radius of orbit. If the radii of 1st, 2nd and 3rd orbits of any atom are r_1, r_2 and r_3 and energies are E_1, E_2, E_3 respectively then, $E_1 < E_2 < E_3$ [Figure 10.1 (a)]. The energy associated with the electrons at different orbits are presented energy level diagram [Figure 10.1 (b)]. Mono atomic material have specific energy shell which are indicated by straight line. But materials made up of by millions atom have some wide energy bands [Figure 10.1 (c)].



10.1 (a) : Orbit 10.1 (b) : Energy level 10.1 (c) : Energy band

Energy Band : There is slightly difference in energy of electrons revolving same orbit in an atom of any matter. The rang between maximum and minimum energy level of these electrons situated at same orbit is called energy band.

Energy level and energy band are not same. Energy level is the orbit, where electrons rotating revolves with specific energy round the nucleus of atoms in matter. Energy band is the energy-range of electrons at same orbit in different atoms of matter.

The main three energy bands of solid are –

(a) Valence Band (b) Conduction Band (c) Forbidden energy Band or gap (Figure 10.2)

(a) Valence Band : Electron at outer most shell of atom is called valence electron. The range of energy of valence electron is called valence band. Valence band in Inert gas is completely filled. The valence band of another matter may be partially or fully filled. This band can never be empty.

(b) Conduction Band : Free valence electron in atom takes part in conducting current. The energy-range of conduction electron is conduction band. All electrons at conduction band are free electrons. In insulator, the conduction band is empty. So, no electric current passes through the conduction band of insulator.

(c) Forbidden Energy Band or Gap : The range of energy between valence band and conduction band is known as forbidden energy gap. No electron can stay in the forbidden energy band because of absence of authorized energy level. In order to transfer an electron from valence band to conduction band, an external energy which is equal to the forbidden energy gap is required.

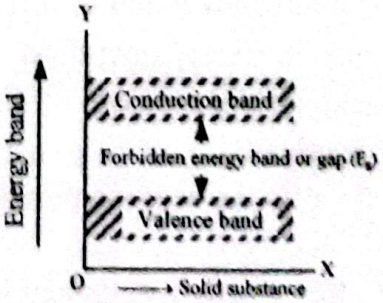


Fig 10.2 : Energy band

A solid can be conductor, semiconductor or non-conductor depending on this gap.

Energy band in solid matter : According to Bohr, electrons can revolve round the nucleus in some authorized orbit of specific energy. Electrons stay in some discrete energy level [Fig. 10.3 (a)]

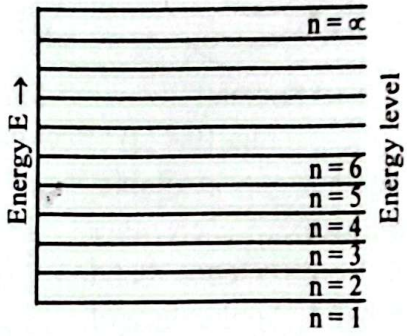


Fig. 10.3 (a) : Energy level of a solid matter.

Atoms are arranged regularly in a solid crystal and they are closed to each other as a result, electrons in different atoms influence to each other and energy level of electrons changes. This change for the electrons at outermost shell is maximum. Because, electrons at the outer most shell of atoms in crystal do not stay under an atom. Electrons at same orbits of different atoms packed closely do not have same energy. As a result, wide energy band is produced through the breaking outer most energy level of atoms in the crystal in solid matter.

Beside electrons at outermost shell of different atoms in crystalline solid matter do not remain isolated. They can easily move to different atoms. Valence band is the range of energy of valence electrons. Again electrons of valence band are almost free. They can move to conduction band attaining some energy. The band between valence and conduction band is called forbidden band.

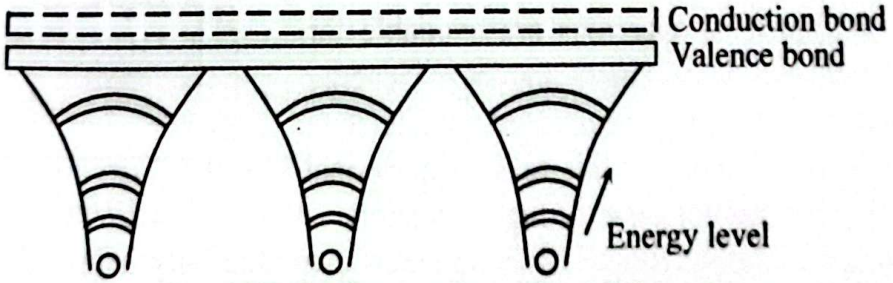


Fig. 10.3 (b) Energy band in solid matter

10.2 Conductor, Non-Conductor and Semiconductor on the Basis of Band Theory

Band Theory : On the basis of forbidden energy gap between valance band and conduction band, solid substances can be classified as conductor, non-conductor and semiconductor. The theory by which the electric conductivity of substance is described is called band theory.

(a) Conductor : The substances through which electricity can Pass easily is called conductors. These substances have 1, 2 or 3 electrons in their outer shell of atom instead of maximum 8 electrons.

It's relative resistance is low approximately 10^{-8} Ωm to 10^{-6} Ωm. All metallic substances such as copper, silver, sodium, magnesium, aluminum, etc. are conductor [Figure 10.4 (a)].

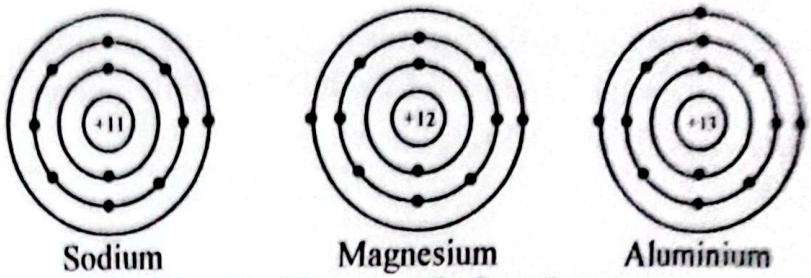


Fig. 10.4 (a) : Bhor model of conductor atom.

In conductors, there is no forbidden gap available, the valence are conduction bond overlap each other [Fig 10.4 (b)]. The electrons from valence band freely enter into the conduction band. Application of very low potential difference can cause the continuous flow of current. If temperature increases, resistance of conductor increases.

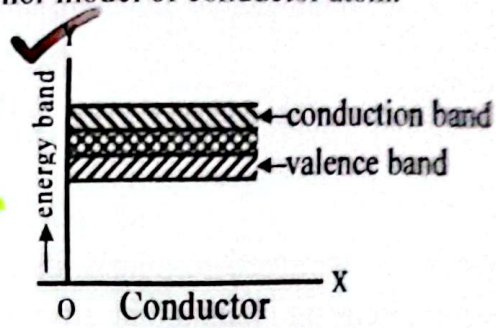


Fig. 10.4 (b) : Conductor.

(b) Non-conductor or Insulator :
The substances through which electricity can not Pass easily is called insulators. These substances have 5, 6 or 7 electrons in their outer shell of atom. Its relative resistance is higher than 10^{13} Ωm. For example, glass, plastic, wood, nitrogen, sulphur, neon etc. are insulators. [Fig 10.5 (a)].

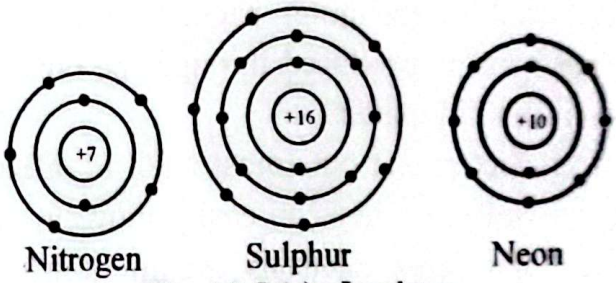


Fig. 10.5 (a) : Insulator.

In an insulator, the forbidden energy gap is very large [Fig 10.5 (b)]. In general, the forbidden energy gap 6 eV is very large and almost no electrons are available for conduction. Therefore a very large amount of energy must be supplied to a valence electron to enable it to move to the conduction band. For this reason, in normal temperature, electricity can not flow through insulators.

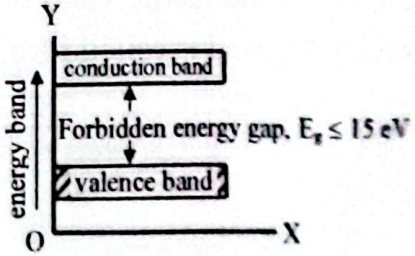


Fig. 10.5 (b) : Insulator.

(c) Semiconductor : The substances whose electric conductivity is less than conductors but greater than insulators and their resistance decreases with the increase of temperature and its electric conduction property changes remarkably due to dropping necessary impurity are called semiconductors.

These substances have exactly 4 electrons in their outer shell of atom instead of maximum 8 electrons. Its relative resistance is approximately 10^{-5} Ωm to 10^{-8} Ωm. Germanium (Ge), Silicon (Si) are pure semiconductors [Fig 10.6 (a)].

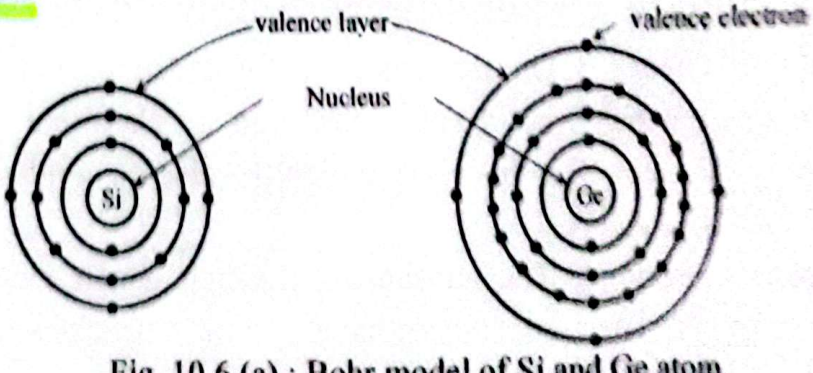


Fig. 10.6 (a) : Bohr model of Si and Ge atom.

In semiconductors, [Fig 10.6 (b)], the forbidden gap is very small. The forbidden energy gap for Ge is 0.7 eV and for Si, it is 1.1 eV. There are no electrons in the conduction band. The valence band is completely filled at 0 K. With a small amount of energy that is supplied, the electrons can easily jump from valence band to the conduction band.

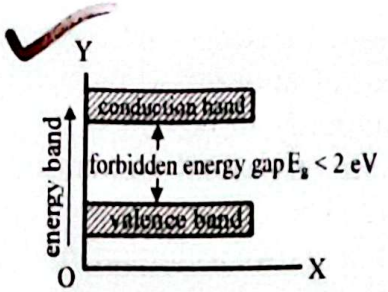


Fig. 10.6 (b) : Semiconductor.

For example, if the temperature increases, the forbidden gap decreases and some electrons liberate into the conduction band. The conductivity of a semiconductor increases.

At low temperature, the valence band of semi conductor is fully filled and conductor band is fully empty. So, it acts as an insulator.

As a result, if a little amount energy is supplied to valence band, electron transfers to conduction band. Again at room temperature, valence band and conduction band are partially filled. With the increase of temperature, valence electron attains energy and gets free. When an electron enters into conduction band from valence band, a hole is created in the valence band.

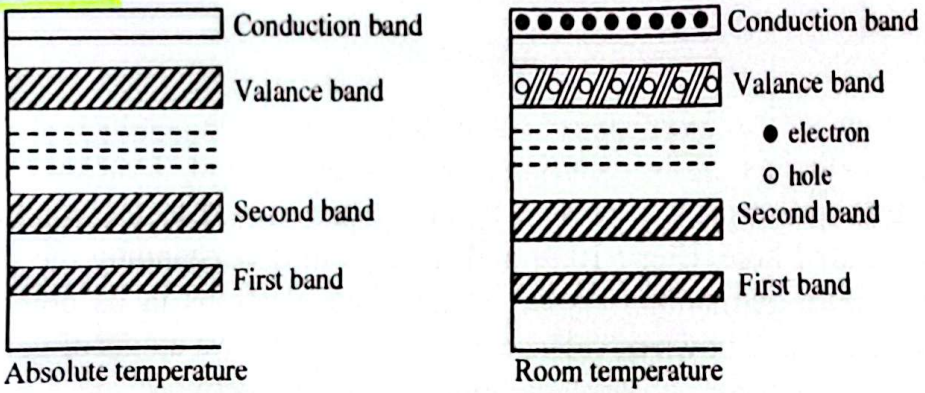


Fig. 10.7 : Energy band at absolute and room temperature.

10.3 Intrinsic and Extrinsic Semiconductor

There are two kinds of semiconductors. Such as —

(a) Intrinsic or pure semiconductors

(b) Extrinsic or impure semiconductors.

(a) **Intrinsic semiconductor :** The semiconductor which contains no impurity is known as intrinsic semiconductor.

Example : The substances which have 4 electrons in their outer shell of atom instead of 8 electrons and belong to 4th group of periodic table are intrinsic semiconductor. Common examples of intrinsic semiconductors are pure germanium and silicon [Fig 10.6 (a)]. Carbon (C_6) is the exception.

(b) **Extrinsic semiconductor :** An extrinsic semiconductor is one in which an impurity is mixed in controlled way to increase the electrical conductivity of the semiconductor. Depending on the types of impurity added, an extrinsic semiconductor can be classified as P-type or N-type.

Doping : Electrons and holes can be generated in a semiconductor crystal adding a small amount of impurity to the semiconductor crystal. As a result, the conductivity increases. The impurity to be added is of the order of 100 ppm (parts per million). The process of addition very small amount of selected impurity to an intrinsic semiconductor is called doping. The impurity atoms are called dopants. The semiconductor containing impurity atoms is known as impure or doped or extrinsic semiconductor.

The element of 3rd row of periodic table Boron (B_5), Aluminum (Al_{13}), Gallium (Ga_{31}) etc. and element of 5th row of periodic table Phosphorus (P_{15}), Arsenic (As_{33}), Antimony (Sb_{51}), Bismath (Bi_{83}) etc. are used for doping.

Characteristic of semiconductor :

1. Relative resistance of semiconductor is 10^{-5} Ωm to 10^{8} Ωm approximately.
2. Semiconductors behave like insulators at absolute zero temperature.
3. Resistance of semiconductor decreases due to increase temperature (Ohms Law is not applicable) so conductivity increases.
4. Energy gap between conduction and valence band is less than 2eV in semiconductor. Such as for Ge it is 0.7eV and for Si it is1.1eV.
5. Electron and hole both are charge carrier in semiconductor.

10.4 Concept of Electron and Hole

The electrons in an intrinsic semiconductor which move to the conduction band at high temperatures are called as intrinsic carriers. In the valence band, a vacancy is created at the place where the electron was present before it had moved to the conduction band. This vacancy is called hole. [Fig : 10.8] and it helps in understanding the creation of a hole. Consider a pure germanium crystal. It has four electrons in its outer or valence orbit. These electrons are known as valence electrons. When two atoms of germanium are brought close to each other, a covalent bond is formed between the atoms. If some additional energy is received, one of the electrons contributing to a covalent bond becomes free to move in the crystal lattice.

While coming out of the band, a hole is created at its place. The hole is usually represented by a open circle. An electron from the neighbouring atom can break the covalent bond and can occupy this hole creating a hole in another place. Since an electron has a unit negative charge, the hole is associated with unit positive charge. The importance of hole is that it may serve as a carrier of electricity in the same manner as the electron but in the opposite direction. Number of electrons and holes are equal in semiconductor.

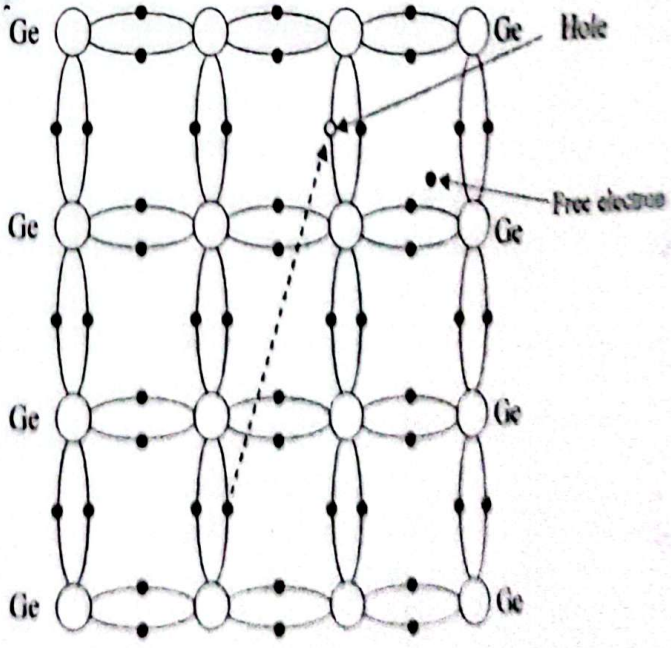


Fig. 10.8 : Free electron and hole.

10.5 p-type and n-type Semiconductor

We know that, extrinsic semiconductor can be, (a) p(positive)-type or (b) n(negative)-type depending on doping.

(a) p-type semiconductor : When a small amount of trivalent impurity (such as indium, boron or gallium) is added to a pure semiconductor crystal, the resulting semiconductor crystal is called p-type semiconductor.

Structure : Fig 10.9 shows a crystal structure which forms, when trivalent boron is added as impurity to pure germanium crystal. The three valence electrons of the boron atom form covalent bonds with valence electrons of three neighbourhood germanium atoms. In the fourth covalent bond, only one valence electron is available from germanium atom and there is deficiency of one electron which is known as a hole. Hence for each boron atom added, one hole is created. Since the holes can accept electrons from neighbourhood, the impurity is called acceptor.

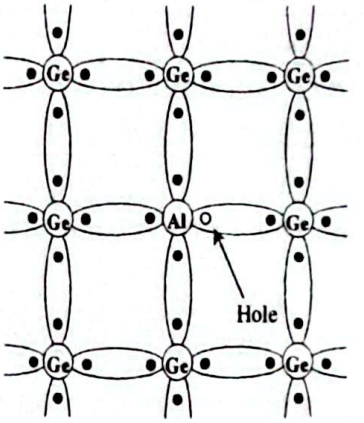


Fig. 10.9 : p-type semiconductor.

The hole may be filled with the electron from a neighbouring atom producing a hole in that position from where the electron moves. This process continues and the hole moves about in a random manner due to thermal effects. Since the hole is associated with a positive charge moving from one position to another, this is called as p-type semiconductors.

In p-type semiconductors, holes are the majority charge carries and free electrons are the minority charge carries.

(b) n-type semiconductor : When a small amount of pantavalent impurity such as arsenic (As_{33}) is added to a tetravalent pure semiconductor such as germanium (Ge_{32}), the resulting crystal is called n-type semiconductor.

Structure : Fig 10.10 shows the crystal structure which forms when pantavalent arsenic is added as impurity to pure germanium crystal. The four valence electrons of arsenic atom form covalent bonds with electrons of neighbouring four germanium atoms. The fifth electron of arsenic atom is loosely bound. This electron can move almost freely as an electron in a conductor moves and hence it will be the carrier of current.

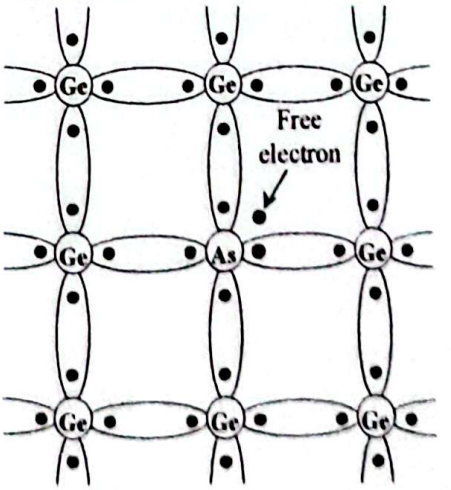


Fig. 10.10 : n-type semiconductor.

When the fifth valence electron is transferred to the conduction band, the arsenic atom become positively charged immobile ion. Each impurity atom donates one free electron to the semiconductor. These impurity atoms are called donors.

In n-type semiconductor material, the number of electrons increases compared to the available number of charge carries in the intrinsic semiconductor. This is because, the available larger number of electrons increases the rate of recombination of electrons with holes. Hence, in n-type semiconductor, free electrons are the majority charge carriers and holes are the minority charge carriers.

Distinction between Intrinsic semiconductors and Extrinsic semiconductor.

Intrinsic Semiconductor	Extrinsic Semiconductor
Intrinsic Semiconductors have no any impurity.	Extrinsic semiconductors contain impurity.
Electric conductivity of intrinsic semiconductor is very low.	Conductivity is greatly increase in extrinsic semiconductor.
The conductivity of intrinsic semiconductor increase with increases of temperature	The conductivity of extrinsic semiconductor depends on amount of atoms added to it.

10.6 Functions of Junction Diode

(a) p-n junction or junction diode : If a p-type and n-type semiconductor are connected to a special arrangement, the connected surface is called p-n junction. A p-n junction is basically made by making one side of a pure semiconductor p-type and another side n type by doping impurities in a controlled way (Fig. 10.11). P region has a high concentration of holes and n region contains a large number of electrons.

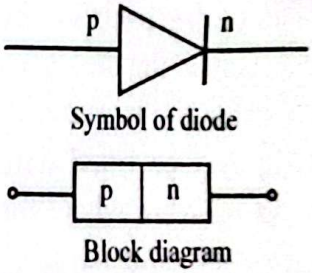


Fig. 10.11 : Junction diode.

(b) Depletion layer and potential barrier : As soon as the junction is formed, free electrons and holes cross through the junction by the process of diffusion. During this process, the electrons crossing the junction from n-region to the p-region, recombine with holes in the p-region very close to the junction. Similarly holes crossing the junction from the p-region to the n-region, recombine with electrons in the n-region very close to the junction. Thus a region is formed, which does not have any mobile charges very close to the junction. This region is called depletion region. n-region becomes positively charged by giving up electrons and p-region becomes negatively charged by accepting electrons. As a result a potential barrier is produced which prevents the flow of electrons from n-region to p-region and stops the current flow after a while (Fig. 10.12). The potential barrier is approximately 0.7 v for a

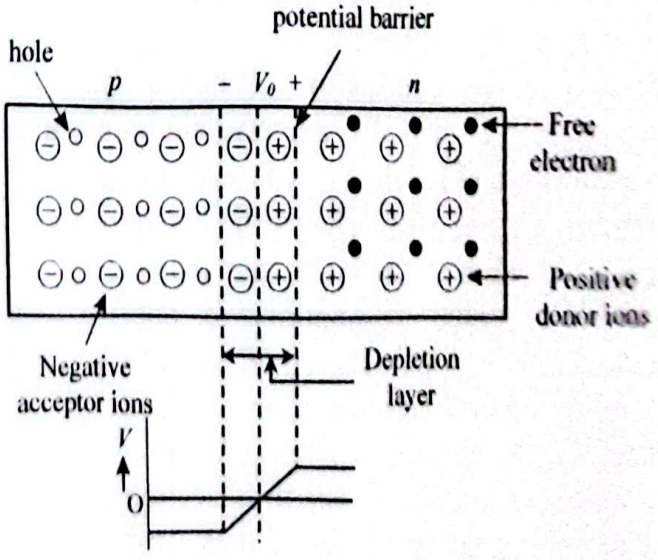


Fig. 10.12 : Depletion layer and potential barrier

silicon p-n junction and 0.3 v for a germanium p-n junction. The distance from one side of the barrier to the other side is called the width of the barrier which depends upon the nature of the material.

Application of external voltage in p-n junction can be two types. Such as–

1. Forward bias and
2. Reverse bias.

1. Forward bias : When the positive and negative terminals of the battery are connected to p-side and the n-side of a p-n junction respectively, the p-n junction diode is said to be forward biased.

When the p-n junction is forward bias (fig 10.13), the applied positive potential repels the holes in the p-region, and the applied negative potential repels the electrons in the n-region. As a result, the charges move towards the junction. If the applied potential difference is more than the potential barrier, some holes and free electrons enter into the depletion region.

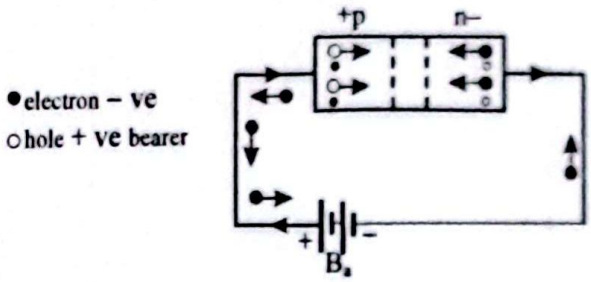


Fig. 10.13 : Forward bias.

Hence, the potential barrier as well as the width of the depletion region are reduced. The positive donor ions and negative acceptor ions within the depletion region regain electrons and holes respectively. As a result, the depletion region disappears and the potential barrier also disappears. Hence, under the action of the forward potential difference, the majority charge carriers flow across the junction in opposite direction and continue current flow in the forward direction.

2. Reverse bias : When the positive and negative terminals of the battery is connected to the n-side and the p-side of a p-n junction respectively the junction is said to be reverse bias.

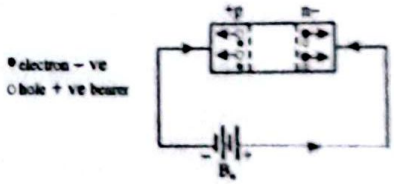


Fig. 10.14 : Reverse bias

When the p-n junction is reverse bias (fig 10.14), electrons in the n-region and holes in the p-region are attracted by the positive and negative terminal of the cell and move away from the junction.

As a result, the number of negative ions in the p-region and positive ions in the n-region increases, Hence the depletion region becomes wider and the potential barrier increases. Since the depletion region does not contain majority charge carriers, it acts like an insulator. Therefore, no current will flow through the external circuit.

But, in fact, a very small current of few microamperes flows in the reverse direction. This is due to the minority carriers flowing in the opposite direction. This reverse current is small, because the number of minority carriers in both region is very small. Since the major source of minority carriers is thermally broken covalent bonds, the reverse current mainly depends on the junction temperature.